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Viewpoint Paper

Thermodynamic modeling of the Mg–Ce–Gd–Y system

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Abstract—An overview is given on ternary phase diagram data involving magnesium with different rare earths (RE), including the Mg–Y–RE systems. An original thermodynamic description of quaternary Mg–Ce–Gd–Y alloys including the ternary subsystems is developed. Similarities and differences of the phase relations involving magnesium with different RE elements and yttrium are highlighted. The relevance for the impact of different components in mischmetal on the properties of Mg–RE alloys is emphasized.
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Keywords: CALPHAD; Phase diagram; Magnesium alloys; Rare earth

1. Introduction

The significance of rare earth (RE) elements in magnesium alloys has been spurred by the demand for high-temperature creep-resistant alloys, especially for automobile powertrain applications, where operating temperatures are in the range of 150–200 °C. Commonly used Mg–Al alloys, such as AZ91 and AM60, do not exhibit sufficient creep resistance at these temperatures. The development of creep-resistant magnesium die-casting alloys in recent years has been focused on additions of alkaline earth metals (Ca, Sr) or rare earth (RE) elements to the Mg–Al system in order to reduce the formation of probably detrimental $Mg_{17}Al_{12}$. Although the formation of eutectic $Mg_{17}Al_{12}$ in the interdendritic or grain boundary region during solidification is suppressed as a result of the preferential reaction of aluminum and RE elements to form $A1_{11}RE_3$ and Al_2RE phases in the benchmark die-cast AE42 alloy, precipitation of $Mg_{17}Al_{12}$ can still occur at elevated temperatures due to the supersaturation of aluminum solute in the (Mg) matrix after die casting [1]. It is therefore desirable to minimize the aluminum content in alloys with dominant RE element additions [2].

Alloying with RE is generally done by adding mischmetal, a mixture of REs with typical composition ranges of La (25–34%), Ce (48–55%), Pr (4–7%), Nd (11–17%) and possibly some Sm. High-grade mischmetal may consist of La (44–50%) and Ce (50–56%) with Pr < 0.5% and Nd < 0.1%. A distinction is made between light RE metals (La, Ce, Pr, Nd, Sm) and heavy RE metals (Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu). Commonly used mischmetal consists mainly of light RE metals. It should be emphasized that the individual RE elements may exert distinctly different effects as demonstrated, for example, by Zhu et al. [3] for the creep properties of two Mg–2.5(Ce, La, Nd)–0.6Zn (wt.%) alloys, one with high La, the other with high Nd content. It is, thus, an oversimplification to lump the RE additions together as a sum of wt.% RE. A first step in understanding the behavior of different RE elements in magnesium alloys is to acquire quantitative knowledge of a consistent description of phase diagrams and thermodynamics, especially beyond the more or less known binary systems. It is the purpose of this work (i) to provide an overview of what is known in this field for Mg–RE–RE systems, including also the element yttrium in Mg–Y–RE systems; (ii) to develop an original thermodynamic description of quaternary Mg–Ce–Gd–Y alloys and its ternary subsystems and to highlight similarities and differences of the phase relations involving magnesium with different RE elements and yttrium.

2. Literature about Mg–RE–RE phase diagrams

There is little literature data about ternary Mg–RE–RE phase equilibria and thermodynamics. Experimental work has been published only for the Mg–Pr–Yb system [4]. Giovannini et al. [4] reported large ternary solubilities of the Pr_5Mg_{41}, $PrMg_3$, PrMg and Mg_2Yb phases

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together with two ternary phases in the isothermal section at 450 °C. No literature is available for any other ternary Mg–RE–RE system. The existence of large ternary solid solubilities of some binary compounds and probably ternary phases may be expected, based on analogy to the scanty information for Mg–Pr–Yb [4] and our own preliminary experimental work on Mg–Ce–Nd alloys.

Mg–Y–RE phase diagrams are better established. For all light RE elements and for Gd experimental investigations of a single isothermal section (mostly at 500 °C) have been published as summarized in Table 1. In addition, CALPHAD-type assessments are available for most of the Mg–Y–(light)RE systems, Mg–Y–La [5], Mg–Y–Ce [6], Mg–Y–Pr [7], Mg–Y–Nd [6,8] and for the only heavy RE example, Mg–Y–Gd [9], producing the entire phase diagram including also the equilibria with liquid phase.

Mg–Y–RE systems typically show large ternary solubilities of the binary intermetallic phases and continuous solid solution of the (RE,Y)Mg phase crystallizing in the CsCl structure in all investigated systems. Truly ternary phases of composition $(RE,Y)_1(Mg)_5$ have been reported in three systems with light RE metals (La, Ce, Pr, Nd). For Sm there are no experimental data in this concentration region. The ternary composition of the C15-type Laves phase $(Ce,Y)_1(Mg)_2$ at 500 °C [10] may not be considered a true ternary phase since it probably originates as solid solution from the binary $CeMg_2$ phase, and is stable only at higher temperatures. For the related binary compound $LaMg_2$ (same C15 structure type, cF24-Cu_2Mg) a solid solution $(La,Y)_1(Mg)_2$ at 500 °C from 0 to 42 wt.% Y is reported [11], see Table 1, whereas the currently accepted binary Mg–La phase diagram [12] shows a eutectoidal decomposition of $LaMg_2$ at 725 °C. However, it is explicitly noted in the ternary study [11] that even after lengthy annealing treatments at 500 °C no significant decomposition was observed, both in the binary $LaMg_2$ and in the ternary $(La,Y)_1(Mg)_2$. The inconsistency concerning the stability of binary $LaMg_2$ at lower temperature is thus unresolved.

Within the Mg–Y–(light)RE systems a trend is visible since the compositions of the binary Mg–RE phases change. The phase $REMg_{12}$ exists only for La, Ce and Pr. Therefore the Mg-rich corner is dominated by the equilibria of $(Mg) + Mg_{24}Y_5 + REMg_{12}$ in the Mg–Y–La, Mg–Y–Ce, Mg–Y–Pr systems compared to $(Mg) + Mg_{24}Y_5 + RE_5Mg_{41}$ in the Mg–Y–Nd and Mg–Y–Sm systems. The ternary Mg–Y–RE phase diagrams with heavy RE (Eu, Gd to Lu) may be quite different. This tentative consideration is indicated by the only known Mg–Y–(heavy)RE phase diagram, i.e. Mg–Y–Gd. Here, the Mg-richest binary Mg–Gd phase is $GdMg_5$ (not RE_5Mg_{41}). The ternary solubilities of the binary phases are remarkably larger for Gd compared to the light RE metals (see Table 1). Another important difference is the absence of ternary phases.

Strictly speaking, IUPAC [13] recommends the inclusion of Sc, Y and the lanthanoids under the name "rare earth metals", where the lanthanoids are (La, Ce, ..., Lu). However, in Mg-alloy designation the letter "E" (synonymous for RE) is used for the lanthanoids only, and in effect simply for mischmetal. Yttrium is considered separately with the letter "W", such as in the popular Mg–Y–RE alloy WE43. This distinction is practical in view of the different effects of Y and RE in Mg alloys, in spite of their similar chemical properties. Scandium is not covered in the present study, even though some work on Mg–Sc [14] and Mg–Al–Sc [15] alloy thermodynamics has been reported and was also applied to the development of creep-resistant Mg–Mn–Sc alloys [16].

3. Thermodynamic modeling of Mg–Ce–Gd–Y alloys

3.1. Binary subsystems

The thermodynamic descriptions of the systems Mg–Y [17] and Ce–Y [18] are taken from the literature. Descriptions of the other four binary subsystems have been refined in the present work, based on literature data, or estimated. The Ce–Mg system is essentially based on Ref. [19], though the (Mg) phase description was improved to obtain a better fitting of the Ce solubility in (Mg) to the experimental data.

Zhang et al. [20] recently reported new experimental findings, indicating that instead of a single $CeMg_3$ phase, two distinct phases with the same cF16-BiF3 crystal structure may exist with compositions $CeMg_3$ and $CeMg_{3.6}$, where $CeMg_{3.6}$ may be a defect–vacancy structure of $CeMg_3$. Moreover, they suggest slightly shifted compositions for the two Mg-richer compounds, namely Ce_5Mg_{39} (instead of Ce_5Mg_{41}) and $CeMg_{11}$ (instead of $CeMg_{12}$). However, in a thermodynamic calculation of the Mg–Ce–Mn system from this same year [21] they have used the very same three Mg-rich binary phases as stoichiometric compounds as we do in the present work, namely $CeMg_{12}$, Ce_5Mg_{41} and $CeMg_3$.

The Gd–Mg modeling is based on Ref. [9]; however, the model of the phase GdMg is simplified in the present work. The experimental evidence for the Gd solubility in solid (Mg) is still scarce and based only on the single experimental work of Rokhlin et al. [22]. The binary systems Gd–Y and Ce–Gd were estimated as ideal solutions of all structurally equivalent phases without any interaction parameter, which reproduces the scarce experimental data [12] in these systems quite well. The software package Pandat [23] was used for all calculations.

3.2. Ternary systems

In this work, we focus on the Mg-rich corner of the ternary systems. Existing thermodynamic calculations of the Mg–Ce–Y system from Ref. [6] and the Mg–Gd–Y system from Ref. [9] could not be used in the present work, since they are based on different assessments for binary subsystems. More importantly, different thermodynamic data for the pure elements were used in these data sets [6,9], namely the new (unpublished) SGTE unary database Version 4.5 from Dinsdale, available in electronic form only. In the present work, we use the published version of the unary database by Dinsdale [24], which is also consistent with our ongoing work on constructing a major thermodynamic Mg alloy database, comprising currently

Table 1. Limit of solid solubilities of the third component in binary compounds in the ternary systems Mg–Y–RE, where Y and the RE element substitute each other. Values are given in wt.% with reference to experimental literature data; entries in parenthesis are calculated values for Ce and Gd from this work at the same temperature.

Phase	Third component	La at 500 °C [11]	Ce at 500 °C [10]	Pr at 500 °C [31]	Nd at 480 °C [8]	Sm at 500 °C [32]	Gd at 500 °C [27]
$Mg_{24}Y_5$	RE	~3	2.7 (2.4)*	2	~4	9	30.7 (33.2)*
Mg_2Y	RE	16	17.2 (15.7)*	16	9	?	58.7 (56.9)*
$REMg_{12}$	Y	~1	3.5 (2.9)*	<1	#	#	#
RE_5Mg_{41}	Y	<1	6.7 (4.7)*	4	5	31	#
$REMg_5$	Y	#	#	#	#	?	22.4 (14.5)*
$REMg_3$	Y	21	22.4 (21.1)*	37	10	?	8.2 (5.9)*
$REMg_2$	Y	42+	12.7–35.5 (30.0–33.8)*	#	#	?	12.6 (8.0)*
T, $(RE,Y)_1(Mg)_5$	Y	11–33	15.8–30.4 (14.4–27.7)*	11.9–35.2	13.3–32.7	?	#

Phase not existing.
? No data available.
* Calculated (this work).
+ Conflicting data on stability of binary $LaMg_2$ are discussed in the text.

20 components and 437 phases. It is noted that, for example, recalculating the binary Ce–Y phase diagram with the binary parameters given in Ref. [6] but using the published unary SGTE data [24] instead of version 4.5 results in a drastically different, and wrong, appearance of this system.

3.2.1. The ternary Mg–Ce–Y system.

The experimental knowledge about this system has been critically reviewed by Flandorfer [25], who also described an experimental investigation of the isothermal section at 500 °C [10]. Continuous solid solution of the (Ce,Y)Mg phase (CsCl-type structure) and large ternary solubilities of the other binary phase were reported (see Table 1). The Mg-rich part of the liquidus surface in given by Drits et al. [26]. They found a eutectic reaction $L = (Mg) + CeMg_{12} + Mg_{24}Y_5$ at 543 °C. Based on these experimental data, thermodynamic parameters for this system were optimized in the present work using the ternary parameters of Ref. [6] as starting values. The calculated isothermal section of the ternary Mg–Ce–Y system at 500 °C is given in Figure 1 and the calculated projection of the liquidus surface in Figure 2. The ternary phase T is modeled as $(Ce,Y)_5(Mg)_2$. The high-temperature phase $CeMg_2$, stable in the binary Mg–Ce system only between 617 and 747 °C, is stabilized at 500 °C in the ternary system by significant exchange of Ce by Y. The Mg_2Y phase is structurally different as discussed later. It is noted that all the $(Ce,Y)_x(Mg)_y$ line compound solutions exhibit a constant atomic composition of Mg; however, since wt.% is used throughout this paper, the straight lines for $(Ce,Y)_x(Mg)_y$ are tilted in the present isothermal sections.

Table 2 shows the calculated invariant reactions of the ternary Mg–Ce–Y system, corresponding to Figure 2. Invariant three-phase reactions such as the minimum in $L + (Ce,Y)Mg + CeMg_2$ at 626.8 °C, the maximum in $L + Mg_{24}Y_5 + T$ at 560.99 °C, and the shallow maximum in $L + Mg_{24}Y_5 + CeMg_{12}$ at 544.99 °C between the two ternary eutectic invariant reactions E_1 and E_2 are not included in Table 2.

The calculated maximum ternary solubilities of the binary phases at 500 °C are shown in Table 1 in parenthesis. Fitting the eutectic temperature of E_2, reported in Ref. [26], together with the relative large solubilities of $CeMg_{12}$ and $Mg_{24}Y_5$, measured in Ref. [10], requires a ternary interaction parameter for the liquid phase. The calculated compositions of the phases at this crucial ternary eutectic E_2 are (in wt.%): 68.0 Mg–12.7Ce–19.3Y (L), 87.5 Mg–0.4Ce–12.1Y (Mg), 62.9 Mg–2.0Ce–35.1Y ($Mg_{24}Y_5$), and 68.8 Mg–28.1Ce–3.1Y ($CeMg_{12}$), respectively.

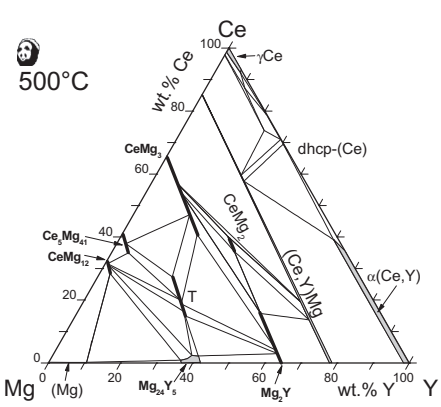

Figure 1. Calculated isothermal section of the ternary Mg–Ce–Y phase diagram at 500 °C. Line compounds, such as Mg_2Y with Ce solubility, are highlighted by bold lines and larger single-phase regions by gray shading.

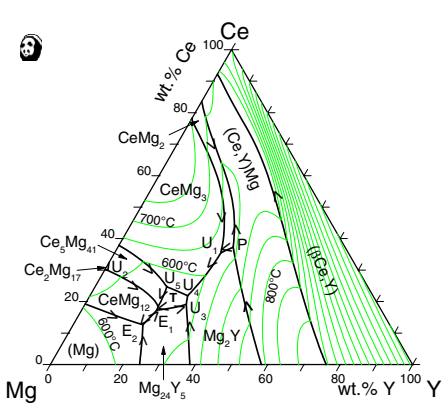

Figure 2. Calculated liquidus surface projection of the ternary Mg–Ce–Y system.

Table 2. Calculated invariant four-phase reactions of the ternary Mg–Ce–Y system involving liquid phase.

Type	T/°C	Reaction
P	631.37	$L + Mg_2Y + (Ce,Y)Mg = CeMg_2$
U1	619.70	$L + CeMg_2 = CeMg_3 + Mg_2Y$
U2	610.14	$L + Ce_2Mg_{17} = CeMg_{12} + Ce_5Mg_{41}$
U3	560.92	$L + Mg_{24}Y_5 = T + Mg_2Y$
U4	556.62	$L + Mg_2Y = T + CeMg_3$
U5	555.39	$L + CeMg_3 = T + Ce_5Mg_{41}$
U6	545.58	$L + Ce_5Mg_{41} = CeMg_{12} + T$
E1	544.34	$L = Mg_{24}Y_5 + CeMg_{12} + T$
E2	543.99*	$L = (Mg) + Mg_{24}Y_5 + CeMg_{12}$

* Confirmed by experimental value, 543 °C [26].

3.2.2. The ternary Mg–Gd–Y system.

The most important work about the ternary Mg–Gd–Y system has been done by Giovannini et al. [27]. They investigated the isothermal section at 500 °C by X-ray powder diffraction, optical and scanning electron microscopy and electron probe microanalysis. Again, a continuous solid solution of the (Gd,Y)Mg phase (CsCl-type structure) and large ternary solubilities of the other binary phase were reported (Table 1). Further experimental investigations were critically reviewed and assessed by Schmid-Fetzer et al. [28] and a ternary invariant reaction involving the phases $Mg_{24}Y_5$ and $GdMg_5$ with (Mg) and liquid below 560 °C is proposed based on the interpretation of measurements reported by Drits et al. [29]. Combining all these experimental data as a basis, the thermodynamic parameters for this system were optimized in the present work, independent of Ref. [9]. The calculated isothermal section of the ternary Mg–Gd–Y system at 500 °C is given in Figure 3, and the calculated projection of the liquidus surface in Figure 4. Table 3 shows the calculated invariant reactions of the ternary Mg–Gd–Y system corresponding to Figure 4. Invariant three-phase reactions close to U_4, such as the minimum in $L + Mg_{24}Y_5 + Mg_2Y$ at 602.66 °C and the maximum in $L + Mg_{24}Y_5 + GdMg_5$ at 605.06 °C are not included in Table 3.

The calculated maximum ternary solubilities of the binary phases are shown in Table 1 in parenthesis. The crucial ternary eutectic E_1 in the Mg-corner involves the following calculated phase compositions (in wt.%):

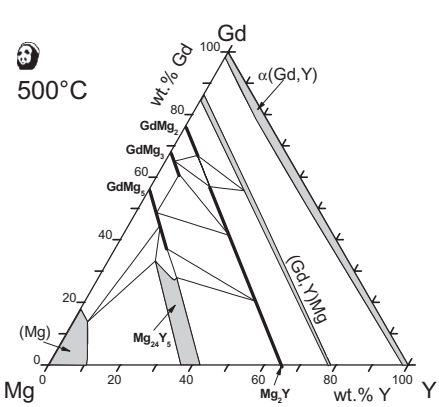

Figure 3. Calculated isothermal section of the ternary Mg–Gd–Y phase diagram at 500 °C.

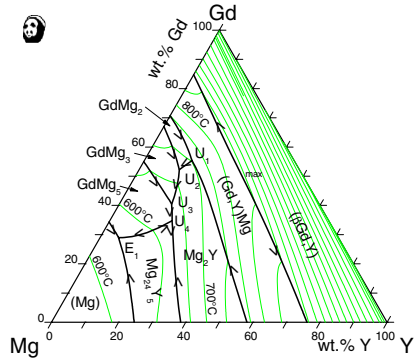

Figure 4. Calculated liquidus surface projection of the ternary Mg–Gd–Y system.

65.5 Mg–28.9Gd–5.6Y (L), 79.7 Mg–16.9Gd–3.4Y (Mg), 52.3 Mg–39.0Gd–8.7Y ($Mg_{24}Y_5$), and 45.6 Mg–48.5Gd–5.9Y ($GdMg_5$), respectively.

3.2.3. The ternary Mg–Ce–Gd system.

No experimental data have been reported for the Mg–Ce–Gd system. In view of the lack of data, no ternary solubilities are modeled in the binary phases $CeMg_{12}$, Ce_5Mg_{41}, $GdMg_5$ and $GdMg_3$. It is obvious, however, that limited ternary solubilities of these binary phases are quite likely, and therefore extrapolative calculations considering these as stoichiometric phases represent only a rough estimation. Experimental work is required before quantitative ternary modeling can be performed, most importantly for the phases $CeMg_{12}$ and $GdMg_5$ which occur in equilibrium with (Mg). The CsCl-type (Ce, Gd)Mg phase and the BiF_3-type $(Ce,Gd)Mg_3$, are estimated as ideal solid solutions of the binary edge phases which is considered a fair approximation for these Mg-poor phases for which tie lines to the Mg-richest phases are very unlikely.

3.3. Estimated quaternary Mg–Ce–Gd-Y phase relations

As an example of possible predictive (extrapolative) calculations, a calculated partial isothermal section of the quaternary Mg–Ce–Gd-Y system at 500 °C and constant 20 wt.% Y is shown in Figure 5. Experimental data for comparison are not available. In view of the restrictions in the ternary Mg–Ce–Gd description, the composition region where both Ce and Gd attain large values is considered uncertain. Nevertheless, the following two key messages can be extracted from the phase diagram outside that range, i.e. at higher Mg-content.

(i) The distinct three-phase equilibrium in the Mg-corner, describing the stable precipitates in hcp (Mg)

Table 3. Calculated invariant four-phase reactions of the ternary Mg–Gd–Y system involving liquid phase.

Type	T/°C	Reaction
U1	700.85	$L + (Gd,Y)Mg = Mg_2Y + GdMg_2$
U2	668.40	$L + GdMg_2 = Mg_2Y + GdMg_3$
U3	611.69	$L + GdMg_3 = GdMg_5 + Mg_2Y$
U4	604.08	$L + GdMg_5 = Mg_{24}Y_5 + Mg_2Y$
E1	550.79*	$L = (Mg) + Mg_{24}Y_5 + GdMg_5$

* Confirmed by experimental value, <560 °C [29].

matrix, is (Mg) + $CeMg_{12}$ + $Mg_{24}Y_5$. These are exactly the same phases as in the ternary Mg–Ce–Y alloys and the added Gd is essentially dissolved in (Mg) and $Mg_{24}Y_5$. These large solid solubilities prevent the occurrence of a separate Gd-based phase, e.g. $GdMg_5$, in that region. It is noted that the phase compositions in this three-phase region are not constant since the tie lines are not in plane with this quaternary section. For example, the composition of the hcp (Mg) matrix is constant only along the ordinate (zero Gd), between 0.5 and 14.3 wt.% Ce with a value of 89.3 Mg–0.2Ce–10.5Y. Upon addition of Gd to the alloy within the (Mg) + $CeMg_{12}$ + $Mg_{24}Y_5$ region in Figure 5, the composition of (Mg) varies up to 84.1 Mg–0.2Ce–9.7Gd–6.0Y at the Gd-rich apex of the triangle. Similarly, the compositions of $Mg_{24}Y_5$ and $CeMg_{12}$ vary.

(ii) Even a single-phase $Mg_{24}Y_5$ region exists with quaternary composition of that phase, emphasizing the significant solid solubilities of both Gd and Ce in $Mg_{24}Y_5$.

The quaternary Mg–Ce–Gd-Y system neatly exemplifies the intricacies to be expected and tackled if extensions to other quaternary Mg–RE–RE–Y systems, or generally Mg–RE multicomponent alloy systems, are to be envisaged. These intricacies mainly concern the extended solid solubilities and the necessity to combine metastable end-member intermetallic phases of different ternary systems into a consistent multicomponent thermodynamic database. In the ideal case structurally equivalent phases should be modeled as a single phase in the thermodynamic description. This is, however, not always a practical approach and decisions have to be made on a case-by-case basis. The following significant solid solution phases have been considered in the example of the quaternary system Mg–Ce–Gd–Y.

3.3.1. Terminal solid solution phases of elemental components.

- *hcp phaseα:* α(Ce, **Y**) and α(**Gd, Y**) probably form a hcp solution α(Ce, **Gd, Y**). Here and in the following bold type indicates major solute components, whereas α(Ce) does not exist as pure element in hcp structure. The (Mg) hcp solid solution is in fact modeled in the current database as the same phase, thus there is only one thermodynamic description for the hcp (**Mg**, Ce, **Gd, Y**) phase; however, it is denoted separately for clarity by (Mg) and α(Ce, **Gd, Y**) in the diagrams. This is because complete mixing of the Mg-rich (Mg) and Mg-poor α(Ce, **Gd, Y**) compositions of this phase is not observed.
- *bcc phase β:* β(**Ce**, **Gd**), β(**Ce**, **Y**) and β(**Gd**, **Y**) probably form a complete bcc solution β(Mg, **Ce**, **Gd**, **Y**) with significant, though limited, solubility of Mg.
- *fcc phase γ:* γ(Ce) dissolves limited amounts of the other components as γ(**Ce**, Gd, Mg, Y).
- *dhcp phase:* dhcp-(Ce), structure type hP4-αLa, dissolves significant, though limited, amount of Y, dhcp-(**Ce**, Y).

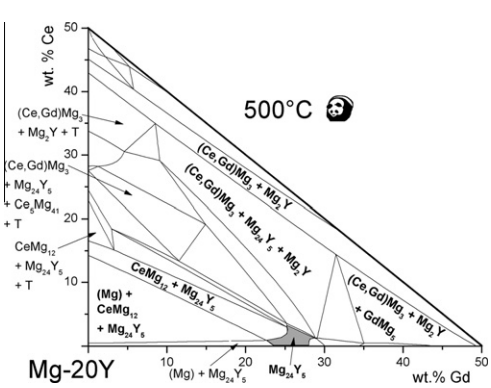

Figure 5. Calculated partial isothermal section of the quaternary Mg–Ce–Gd–Y system at 500 °C and constant 20 wt.% Y for alloys with > 30 wt.% Mg.

3.3.2. Intermetallic phases, originating as binary compounds.

- *$(RE/Y)_1Mg_1$ phase:* (**Ce,Y**)Mg and (**Gd,Y**)Mg probably form a cP2-CsCl-type solution, modeled as single-phase (**Ce, Gd, Y**)Mg.
- *$(RE/Y)_1Mg_2$ phases:* (**Ce**, Y)Mg_2 and (**Gd**,Y)Mg_2 probably form a cF24-Cu_2Mg Laves-type solution (**Ce**, **Gd**, Y)Mg_2. In this work it was modeled for simplicity as two separate phases, since the temperature dependence of the solubility limit of (**Gd**,Y)Mg_2 with the high-temperature phase (**Ce**, Y)Mg_2, stable only over a small temperature range (747–617 °C) in Mg–Ce, is not clear in the ternary Mg–Ce–Gd, not to speak of the quaternary system, where this phase is stabilized by substitution of Ce with Y.
- The Mg_2Y phase with similar composition 2:1 crystallizes in different Laves-type, hP12-$MgZn_2$, and must therefore be modeled as separate phase, (Ce,Gd,**Y**)Mg_2.
- *$(RE/Y)_1Mg_3$ phase:* (**Ce**, Y)Mg_3 and (**Gd**,Y)Mg_3 probably form a cF16-BiF_3-type solution, modeled as single-phase (**Ce**, **Gd**, Y)Mg_3.

3.3.3. Intermetallic phases with Mg-content higher than *$(RE/Y)_1Mg_3$*.

- *$Mg_{24}Y_5$ phase:* Structure type cI58-αMn, Mg_{24}(**Y**, Ce, Gd, Mg)$_5$ dissolves not only some Ce and substantial amounts of Gd but also some Mg as binary non-stoichiometric Mg–Y phase.
- *$GdMg_5$ phase:* Structure type cF440-$GdMg_5$, (**Gd**, Y)Mg_5 dissolves some Y.
- *Ce_5Mg_{41} phase:* Structure type tI92-Ce_5Mg_{41}, (**Ce**, Y)$_5$**Mg_{41}** dissolves some Y.
- *$CeMg_{12}$ phase:* Structure type tI26-$ThMn_{12}$, (**Ce**, Y)**Mg_{12}** dissolves some Y.

Only the binary Ce_2Mg_{17} compound, stable over only a small temperature range (617–609 °C) is considered here a "simple" stoichiometric phase without any solid solubility in the Mg–Ce–Gd–Y system.

4. Conclusion and outlook

Published data on Mg–RE–RE phase equilibria are scarce [4]. More is known when the scope is extended to yttrium in Mg–Y–RE systems. Even thermodynamic CALPHAD-type assessments are available for most of the Mg–Y–(light)RE systems, except for Sm. This provides a first approach to the effects of alloying com-

monly used mischmetal, which consists mainly of light RE metals. The effects of heavy RE are known only for the phase diagram Mg–Y–Gd.

Developing consistent descriptions of phase diagrams and thermodynamics for Mg–RE–RE and Mg–Y–RE systems has been exemplified here for quaternary Mg–Ce–Gd–Y alloys and its ternary subsystems. The intricacies to be tackled for such Mg–RE–RE'–Y multicomponent alloy systems mainly concern the extended solid solubilities of intermetallic phases but also of the terminal solution phases. For quaternary and higher-order systems, the proper consideration of structurally equivalent phases and the necessity to combine metastable end-member intermetallic phases of different ternary systems into a consistent multicomponent database is a major challenge.

It is emphasized that the individual RE elements may exert distinctly different effects on the phase relations and, thus, it is an oversimplification to lump the RE additions together as a sum of wt.% RE. This distinction is also observed in the effect on mechanical properties as demonstrated, for example, by Zhu et al. [3].

Acquiring quantitative knowledge of a consistent description of phase diagrams and thermodynamics for Mg–RE–RE'–Y multicomponent alloy systems is a major task for future research. The focus for application to magnesium alloys should be on experimental determination of the crucial ternary solid solubility of intermetallic phases, at least for those which might form equilibria with (Mg) at some temperature. Determination of the pertinent invariant reactions involving the liquid phase and the melting equilibria and enthalpies of ternary phases are also most important. Even for binary Mg–RE systems, selected experimentation is required, at least in case of less-established (Mg) solvus and the invariant Mg-rich reactions. Consistent thermodynamic CALPHAD-type assessments have to be generated based on such data, considering the intricacies detailed above. Such work is in progress for the Mg–Ce–Nd system. Finally, such generation of thermodynamic descriptions needs to be fully consistent with other alloying components (Al, Zn, Mn, etc.) in order to provide a tool for calculating the quantitative impact of different REs for a broader range of magnesium alloys. Especially for zinc, quite complex multicomponent phase relations are anticipated, e.g. in Mg–Ce–Zn [30], with extended solid solubilities involving also Zn in the intermetallic phases.

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